



KLE Technological
University
Creating Value
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**Department of
Electronics and Communication Engineering**

**Mini Project Report on
MOBILE PHONE CHARGER - IC REGULATOR BASED
DESIGN (5V, 1A)**

By:

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K.L.E SOCIETY'S
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2025-26



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the project entitled "Design and Implementation of a Regulated 5V, 1A DC Power Supply from 230V AC" is a bonafide work carried out by the student team of "Soubhagya Hundekar(01FE23BEC272), Shivani Solapur(01FE23BEC274), Srushti Rajput(01FE23BEC275), Khushi Tavakari(01FE23BEC280)". The project report has been approved as it satisfies the requirements with respect to the Mini project work pre-scribed by the university curriculum for BE (V semester) in Department of Electronics and Communication Engineering of KLE Technological University for the academic year 2025- 26.

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We would like to recognize the unity and concerted effort of the entire team members who were actively involved and engaged in extensive discussions and collaboration that helped to complete this project successfully.

-Team E5

ABSTRACT

The mini project involves developing a design simulation and analysis work for a 5V 1A IC-powered DC power supply system which provides mobile charging capabilities. The project purpose is to convert 230V AC mains power into stable 5V DC power which will supply energy to mobile devices and low-voltage electronic equipment. The proposed system solutions utilize a step-down transformer together with a full-wave bridge rectifier and filter circuit and IC-based voltage regulation stage to deliver power.

The power supply system uses the LM323T voltage regulator IC as its power supply component because of its ability to deliver high output current while maintaining precise voltage control. The rectifier stage transforms AC voltage into pulsating DC while the high-value filter capacitor is used to eliminate ripple content and produce a continuous output signal. The LM323T regulator system maintains a fixed output voltage of 5V despite changing input voltage and load current conditions. The external capacitors improve stability and noise reduction while the IC uses current limiting and thermal protection features to ensure safety and dependable operation.

The complete circuit design was realized through Proteus software which demonstrated that voltage control remained stable while the system produced lower ripple and maintained steady performance across different load scenarios. The proposed IC-regulated power supply design achieves both design simplicity and operational safety which makes it suitable for mobile charging applications and other low-power electrical needs.

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Chapter 1 Introduction

The need to have a stable and reliable DC power supply is critical in modern electronic systems. The majority of electronic equipment and in particular mobile phones use low-voltage DC power, but the main power that is available is high voltage AC. Therefore, there is a need to have an effective power conversion system that transforms AC power to regulated DC power. Power supplies are also very important in safeguarding electronic devices under fluctuation of voltages, electrical noise and overcurrent conditions. The results of poorly controlled power supply include overheating, shorter battery. In this mini project, the proposed project aims at the design and simulation of a 5V, 1A regulated DC power supply to be used in charging different mobile devices based on the linear regulation methods. A voltage regulation method based on zener diode regulation and a voltage regulation based on LM323T is also implemented, simulated, and compared in order to examine the performance, efficiency, and the suitability of these two methods to real-life users.

1.1 Motivation

With the growing rate of adoption of mobile phones and portable electronic gadgets, there has been a high demand for safe, efficient and reliable methods of charging the gadgets. The power supply at domestic houses is usually prone to voltage variations, and when not regulated well, this may negatively impact electronic appliances. The project will also serve to close the gap between the theoretical and practical design of power supply systems through simulation tools to examine the performance and efficiency and thermal characteristics of the system prior to hardware implementation. We have observed that low-voltage electronics, such as the small chargers, embedded systems, and development boards, have been increasingly emerging, and everybody needs a reliable and stable 5V supply. Old-fashioned voltage regulators simply fail to meet modern performance and safety requirements, and therefore IC based voltage regulators are becoming the new standard.

1.2 Objectives

- Design and Simulation: Create a functional model of a 5V, 1A DC power supply tailored for mobile charging applications.
- AC-to-DC Conversion: Investigate the efficiency and mechanics of rectification (converting AC to pulsating DC) and filtering (smoothing the signal).
- Voltage Regulation: Implement and evaluate IC-based regulation techniques to maintain a constant output despite input fluctuations.
- Load Stability Analysis: Analyze how the output voltage behaves under different loading conditions to ensure consistent charging.

- Ripple Voltage Evaluation: Measure the effectiveness of the filter stage by calculating the residual AC ripple present in the DC output.
- Current-Handling Performance: Study the circuit's ability to provide a steady 1A current without voltage sag or component failure.
- Thermal & Power Analysis: Examine the power dissipation and thermal characteristics of linear regulators to determine if heat sinking is required.
- Comparative Study: Compare two different regulation methods (e.g., a standard 7805 vs. an adjustable or LDO regulator) to justify the final component choice based on performance data.

1.3 Problem statement

The design of a controlled DC power supply that can convert 220V/230V AC main supply into a reliable 5V DC output with a current of 1A and charged by a mobile device is the issue tackled in this project. The system has to be able to give a constant output voltage regardless of changes in input voltage and load current. It must minimize ripple errors and power loss, and provide safe and reliable high performance. The design must also be affordable , simple to install and it must have the ability to run continuously without excessive heating.

1.4 Organization of the report

To introduce the project in a systematic way, this report is divided into seven chapters.

chapter 1, gives an introduction of the project where motivation , objectives and problem statements are introduced.

In chapter 2, review of linear voltage regulators, study of ripple reduction and thermal management are discussed.

In chapter 3, market survey details are discussed.

In chapter 4, the electronic system design and other alternatives of voltage regulation methods are discussed.

In chapter 5, the implementation details and system specification and how the system will operate are explained.

The chapter 6, contains the results and discussions of the simulation results in the context of the performance analysis.

Chapter 7, is the closure of the project and indicates future scope and social applications.

The chapters are progressive in terms of theoretical points to practical assessment.

Chapter 2

Literature Survey

2.1 Review of Linear Voltage Regulators

A fundamental topic in electronics, linearly regulated power supplies continue to be commonly employed in applications that require low power and minimal noise. By minimizing high-frequency switching and utilizing a series pass element controlled by a feedback loop, they generate a stable DC output from a higher DC input, all while ensuring remarkably low output noise and ripple.

The conventional fixed 5 V regulator commonly used in laboratory and hobbyist applications is the 78XX series, especially the 7805 model, which is capable of delivering between 1 to 1.5 A when adequately heat-sinked. These regulators require only a limited number of external capacitors and integrate a reference, an error amplifier, a series transistor, and fundamental protection mechanisms.

Devices like the LM323T are employed for applications requiring higher currents. The LM323T is a three-terminal 5 V voltage regulator that can provide up to approximately 3 A, given adequate cooling, by utilizing a larger pass transistor, internal current limiting, and thermal shutdown features. Utilizing a 3 A-rated device with a 5 V, 1 A charger enhances long-term reliability and provides a reassuring safety margin.

2.2 Study of Ripple Reduction Techniques

A filter capacitor is incorporated into a full-wave bridge rectifier to generate a more stable DC output, as the rectifier's output exhibits significant ripple. The required capacitance for a full-wave rectifier is often determined using the formula

$$C \approx I / 2fV_{\text{ripple}}$$

A single large electrolytic capacitor, or several connected in parallel, is often discussed in power supply design literature, sometimes alongside smaller ceramic capacitors. Small ceramic capacitors are effective at diminishing high-frequency components, whereas larger electrolytic capacitors are designed for substantial energy storage. Adequate input capacitance is essential for regulators like the LM323T to ensure that the input voltage remains above the dropout threshold, even during the lowest point of ripple. Failing to achieve this can result in the regulator briefly falling out of regulation.

2.3 Thermal Management Studies

With power approximately $P_{diss} \approx (V_{in} - V_{out}) \times I_{load}$, the voltage difference between the input and output in linear regulators is converted into heat within the pass device.

If thermal design is overlooked, the junction temperature could exceed safe limits as this product operates.

Research on TO-220 and TO-3 packages indicates that the junction temperature is primarily influenced by power dissipation and the overall thermal resistance between the junction and the surrounding environment. Datasheets for regulators such as the LM323T present derating curves and maximum dissipation ratings, highlighting the conditions under which heatsinks become necessary. The implementation of appropriate heatsinks, effective mechanical contact—often facilitated by thermal paste—and a design that strategically positions hot components away from sensitive circuitry are all sound recommendations. This becomes especially vital when supplying around 1 A at 5 V from a significantly higher input.

Chapter 3

Market Survey

3.1 Commercial Solutions

The modern commercial mobile chargers are mostly typically high frequency SMPS topology. The branded adapters (Apple, Samsung, etc.) are small, efficient (usually more than 85-90%), and can be used in several voltage modes (or in a fast-charging standard such as USB Power Delivery).

Less efficient and poorer safety margin designs are sometimes employed to make cheaper local chargers. They can have more ripple, worse isolation and less dependable short-circuit or overload protection which can impact on battery life as well as safety to users.

The charger of this project is a 5 V, 1 A, linear, transformer-based charger based on LM323T. It is heavier and less efficient than SMPS instruments but has a very simple organization and low noise output, so it is very well adapted as an educational design as well as a clean supply to small embedded systems.

3.2 Cost Analysis

A naive cost analysis is a comparison of the student prototype and the commercial ones. Local retail prices may be: transformer 150 INR, LM323T 45 INR, PCB and capacitors 60 INR, rectifier and connectors 40 INR, heatsink 30 INR and potential enclosure 80 INR. Overall, one unit would cost around 300-350 INR, and it would be dependent on object quality and mechanical finishing.

When bought in large quantities, the prices of components will become quite lower, however a transformer-based linear charger remains more costly and heavier than an SMPS-based adapter with the identical rating. This means that linear chargers are rare in the consumer market due to this cost-size handicap, although they are still useful in a laboratory and special- purpose equipment.

3.3 Safety Standards

The commercial mobile chargers should also meet national and international safety standards.

In India, the Bureau of Indian Standard (BIS) mandates the specifications of insulation, creepages, and clearance distances, temperature increment, leakage current, and marking, and provided certified products bear the BIS mark. Most of the other markets feature CE and similar marks as signs that an electrical safety or electromagnetic compatibility regulation is met.

An approved charger should be able to endure abnormal conditions like overloads, short circuiting, surges and high ambient temperatures without igniting fire or exposing a user to electric shock. Although the charger in this project has been based on the LM323T, which is only an academic prototype, not a certified product, the knowledge of these standards affects design decisions such as isolating the transformer, providing fuses and suitable earthing where needed, and safe enclosure and PCB arrangement.

Electronic System design

4.1 Alternative Solutions

To achieve regulation over a low-voltage DC supply of power, then, the most common options are either taking a Zener diode shunt regulator or snatching an integrated linear regulator IC. Using a series resistor and Zener, you just set the Zener and series resistor to a voltage to clamp the voltage, which only works up to 100200mA of load current and you tend to observe very poor load regulation and efficiency. Conversely, the IC regulators such as the LM323T are capable of about 1.5-3A, are much more precise in regulation, include internal protection, and, again, you do not need to overwork at thermal regulation. That is why, in the case of a mobile charger of 5V, 2A, we decided on the choice of the linear regulator LM323T.

4.2 Functional block diagram

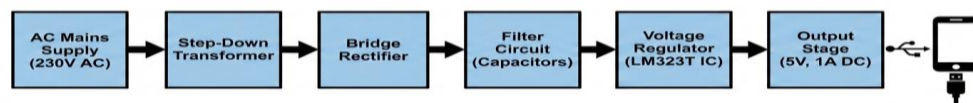


Fig: Functional Block Diagram

Stage 1: AC Mains Supply

The system is powered from the standard 230 V AC mains supply. This high-voltage AC input serves as the primary source of electrical energy for the charger circuit. Since electronic components cannot operate directly at this voltage level, further processing is required.

Stage 2: Step-Down Transformer

The AC mains voltage is applied to a step-down transformer, which reduces the voltage from 230 V AC to a lower AC voltage suitable for the charging circuit. In addition to voltage reduction, the transformer provides electrical isolation, ensuring safety for both the circuit and the connected mobile device.

Stage 3: Bridge Rectifier

The low-voltage AC output from the transformer is fed into a bridge rectifier. This stage converts the alternating current into pulsating DC by using four diodes arranged in a bridge configuration. The rectifier allows current to flow in the same direction during both positive and negative half cycles of the AC input.

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Stage 4: Filter Circuit

The rectified DC output contains ripples and fluctuations. To reduce these ripples, a filter circuit using capacitors is employed. The filter smoothens the pulsating DC into a more stable DC voltage by charging during peak voltages and discharging during voltage drops.

Stage 5: Voltage Regulator (LM323 IC)

The filtered DC voltage is applied to the LM323 voltage regulator IC, which regulates the

output to a constant 5 V DC. This stage ensures voltage stability even when there are variations in input voltage or load conditions. The regulator also protects the mobile device by preventing overvoltage.

Stage 6: Output Load (Mobile Phone)

The regulated 5 V DC output is supplied to the USB output terminal, to which the mobile phone is connected. The circuit delivers sufficient current to charge the mobile phone safely, confirming the proper operation of the charger.

Chapter 5

Implementation details

This chapter describes the process of practical designing and implementation details of mobile phone charger by utilizing an IC based voltage regulator. The design choices constructed in this chapter have been guided by the comparative analysis involving the Zener diode regulation and the IC regulation analyzed above at a detailed level. The emphasis is laid on the compliance with modern charger demands like stability of output voltage, the increased current capacity, and the safety of working.

5.1 Specifications and system Design

The charger has been designed to deliver a controlled DC output appropriate to charge a mobile phone and any other low-voltage electronic device. Given the current power needs, it is decided to take an IC-based linear voltage regulator as the core element of the system.

Design Approach:

Based on the research conducted within the project, the Zener diode regulation does not fit in the much needed criteria of the modern mobile chargers because of poor load control, large power dissipations, and lack of protection features. Thus, an IC voltage regulator is employed which relies on a feedbacking mechanism and can act as a constant output voltage even when the input voltage or the load current varies.

The system design comprises of the following steps:

1. Input Specifications

- Input Voltage: 230V AC $\pm 10\%$, 50 Hz
- Input Type: Single-phase mains supply

2. Transformer Specifications

- Type: Step-down transformer
- Primary: 230 V AC
- Secondary: 12V AC RMS

3. Rectifier Stage Specification

- Rectifier Type : Full-Wave rectifier
- Diodes used : 1N5408 (bridge configuration)
- Maximum Forward Current : 3A (data sheet)
- Peak Inverse Voltage(PIV) Required :1000V
 $V_{in}(\text{rectifier}), V_{peak} = 12 * 1.414 = 16.97 \text{ V}$

4. Rectified DC Voltage

- Secondary RMS voltage: $V_{rms} = 12 \text{ V}$
- Peak voltage: $V_{peak} = 12 * 1.414$

- Voltage drop across diode = 2V (1V for each cycle)
- Output of rectifier, $V_{dc}(\text{peak}) = 16.97 - 2$
 $= 14.97 \text{ V} \sim 15 \text{ V}$

5. Filter capacitors

- C1 (main filter) : 4700 $\mu\text{F}/35 \text{ V}$
 Let load current = 1A
 $C = \text{load current} / (2 * \text{Frequency})$
 $= 1 / (2 * 100)$
 $= 5000 \mu \text{ F}$
- Function : Reduces ripple from rectifier output
- Optional capacitors at the input and output of the regulator: $C_{in} = 1 \mu\text{F}$
 $C_{out} = 0.1 \mu\text{F}$

6. Voltage Regulator(LM323T)

- Output Voltage : 5V DC
- Maximum output current : 1-3 A
- Internal Features: Thermal Shutdown, current limiting, short-circuit protection
- Recommended: Heat sink for high current

7. Power Dissipation in Regulator

- Power dissipation : 10W
- Input to regulator : 15V

$$P = (V_{in} - V_{out}) * \text{current}$$

$$= (15 - 5) * 1 \text{ A}$$

$$= 10 \text{ W}$$

8. Output Specifications

- Output voltage : 5V DC regulated
- Output Current capacity : 1 to 3 A
- Ripple at output : Very low(few milli volts)
- Line and load regulation : Good for small variations

Thermal Considerations:

As linear voltage regulators are power dissipaters due to heat generation, thermal performance is of great concern. The dissipation of power is also determined by the voltage difference between the inputs and outputs and the load current. To provide safe operation in terms of constant use, sufficient heat sinking is given to the regulator in order to maintain the temperature within safe limits.

Generally, the IC-based design has a stable voltage, increased current carrying capacity, and is safer than the Zener diode regulator.

5.2 Comparative Analysis of LM323T and 7805 Voltage Regulator ICs

Parameter	LM323T IC	7805 IC
Type	Fixed positive voltage regulator	Fixed positive voltage regulator
Input Voltage Range	9V - 20V	7V - 25V
Output Voltage	5V	5V
Maximum Output Current	Up to 3A	Up to 1A
Power Dissipation Capability	High	Moderate
Thermal Performance	Better(designed for high current)	Limited at high current
Efficiency	Higher at high current	Lower at high current
Heat Sink Requirement	Mandatory for high current	Required above ~500mA
Cost	Higher	Lower

5.3 Comparative Analysis of 1N5408 and 1N4001 Rectifier Diodes:

Parameter	1N5408	1N4001
Diode Type	General purpose rectifier diode	General purpose rectifier diode
Maximum Avg. Forward Current	3A	1A
Forward Voltage Drop	0.8 - 1.0V	0.7 - 1.0V
Power Handling	High	Low
Peak Repetitive Reverse Voltage	1000V	50V
Surge Current Capability	High	Low
Thermal Capacity	Higher	Lower
Reliability at High Current	Very good	Poor
Suitability for 1A Power Supply	Highly suitable	Not recommended
Typical Applications	Power supplies, rectifiers, battery chargers	Low-current rectifiers

5.4 Flowchart

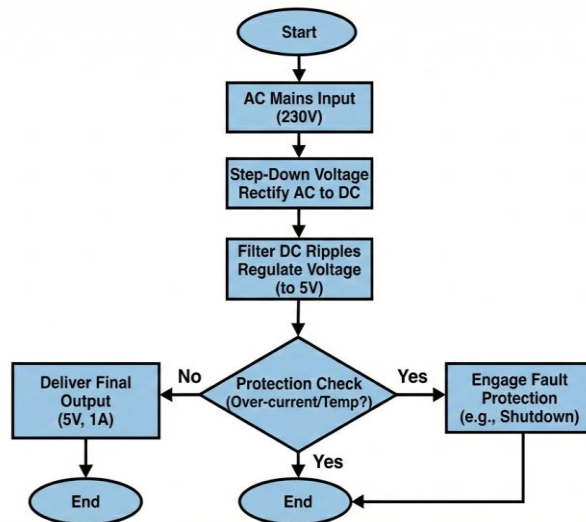


Fig: Flow Chart

Start

The process begins when the system is powered ON and the AC mains input (230 V) is applied to the circuit.

Step 1: AC Mains Input

The AC supply is taken as the input source for the charger. This initiates the power conversion process required for mobile phone charging.

Step 2: Step-Down Voltage

The high-voltage AC input is reduced to a lower AC voltage using a step-down transformer,

making it suitable for further processing and ensuring safety.

Step 3: Rectification (AC to DC Conversion)

The reduced AC voltage is converted into DC voltage using a rectifier circuit. This DC output is pulsating in nature and cannot be used directly for charging.

Step 4: Filtering and Voltage Regulation

The pulsating DC voltage is passed through a filter circuit to reduce ripples. After filtering, a voltage regulator regulates the voltage to a fixed 5 V DC, which is required for mobile phone charging.

Step 5: Protection Check (Over-current / Temperature)

Before delivering the output, the system checks for fault conditions such as over-current or excessive temperature.

If a fault is detected (Yes):

The system activates fault protection (for example, shutdown or current limiting) to prevent damage to the circuit and the connected mobile device. The process then terminates safely.

If no fault is detected (No):

The system proceeds to deliver the output voltage.

Step 6: Deliver Final Output

A regulated output of 5 V, 1 A DC is supplied to the load (mobile phone). The mobile phone starts charging, indicating correct operation of the circuit.

End

Once the output is delivered successfully or a protection condition is triggered, the process ends.

Chapter 6

Results and discussions

6.1 Simulation Analysis

Having simulated the LM323T in Proteus, the test circuit output voltage remained relatively near 5V under all of the input and load conditions used. The load regulation when not loaded to approximately 1.5-2A was within a range of about 0.3-0.5 that is good load regulation in a linear regulator. In adjusting the input to the transformer about the nominal output the curve of the line regulation changed the output voltage hardly at all, and the LM323T clearly maintains that 5V constant.

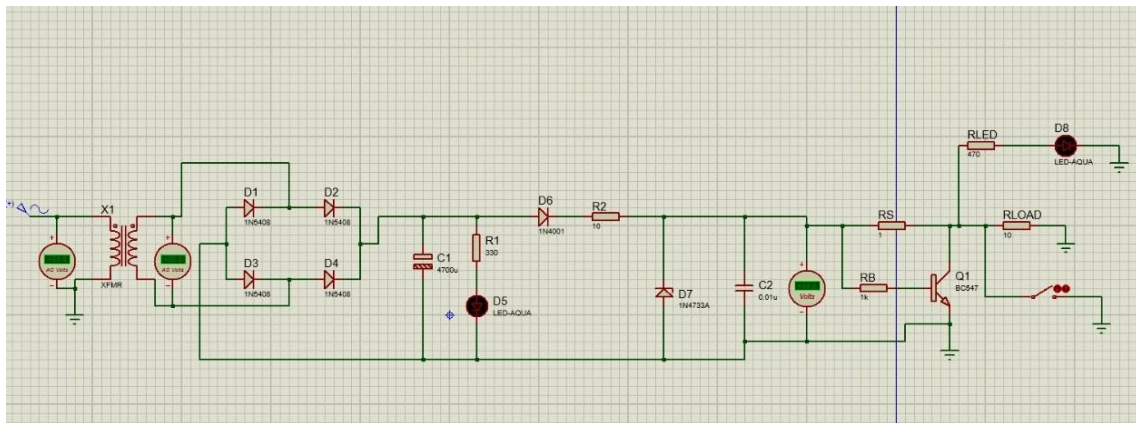


Fig: Simulation Circuit of Zener Diode

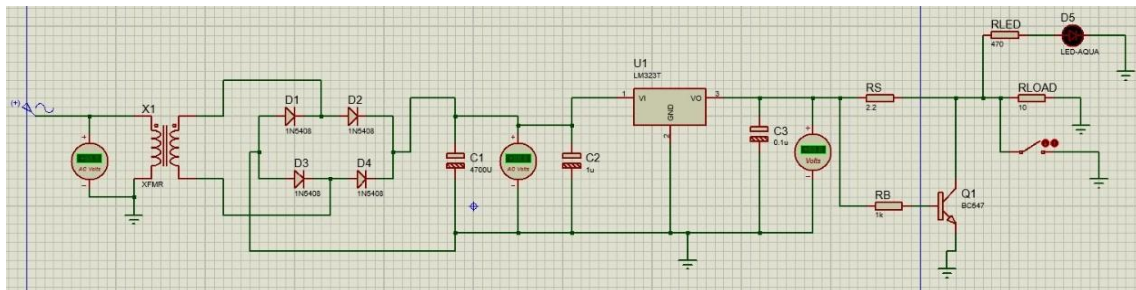


Fig: Simulation Circuit of LM323T

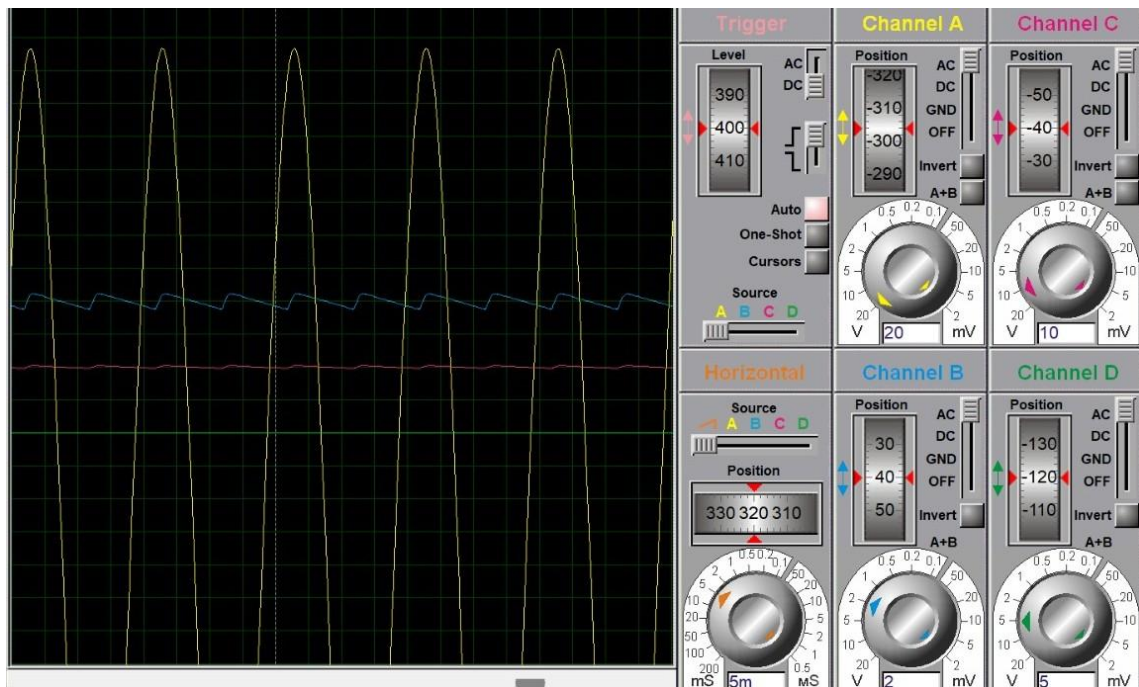


Fig: Simulated Output Waveforms

channel A is ac input, channel B is after rectification, channel C is after capacitor filter, channel d is after regulator IC

6.2 Hardware Realization

The circuit was constructed on a hardware board with the transformer, bridge rectifier, filter capacitor and an LM323T regulator and a heatsink. The no-load output voltage was brought to around 5V and the voltage was loaded and only reduced slightly to provide the rated current. The regulator became hot at higher currents, but it remained in safe temperatures due to the heatsink with the internal thermal protection. On the whole, the lab results were consistent with those we observed in the simulations.

6.3 PCB Results

The charger circuit was realized on a one-layer PCB with requirements of small size and stable functioning. The PCB was tested by plugging the mobile phone into the USB output directly.

PCB performance in the charging process:

- The mobile phone was identified instantaneously and it began charging continuously.
- There was not too much heating to the PCB tracks or components.
- The voltage on the PCB output terminals was steady with continuous charging.
- The IC regulator was safely running with adequate heat dissipation.

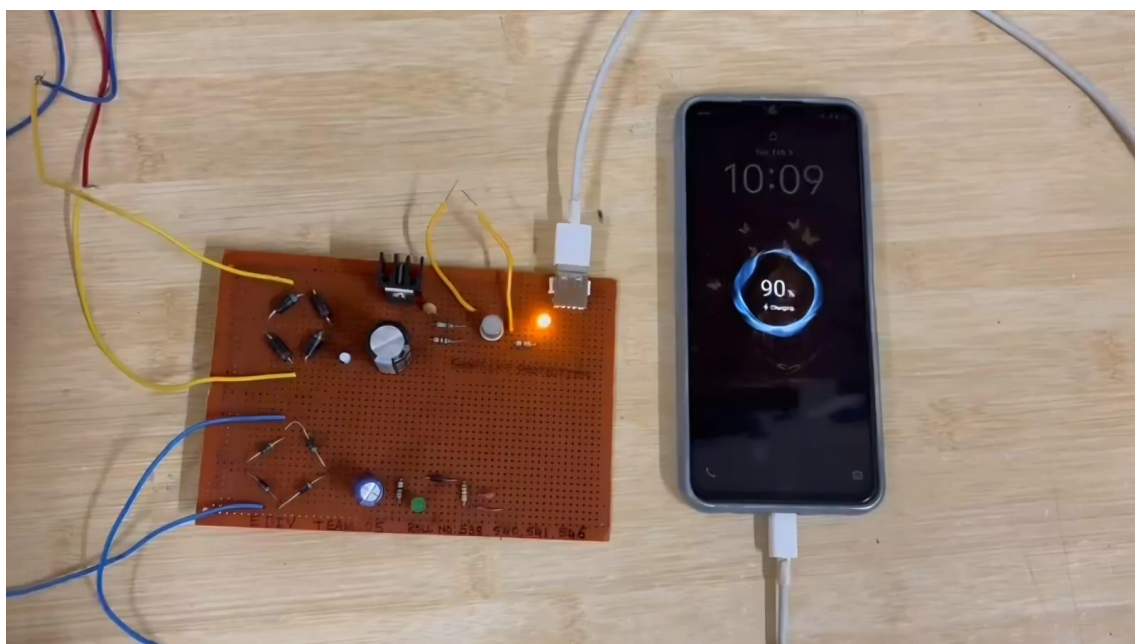


Fig: Hardware implementation of IC-based mobile phone charger indicating successful charging of a mobile phone with the PCB.

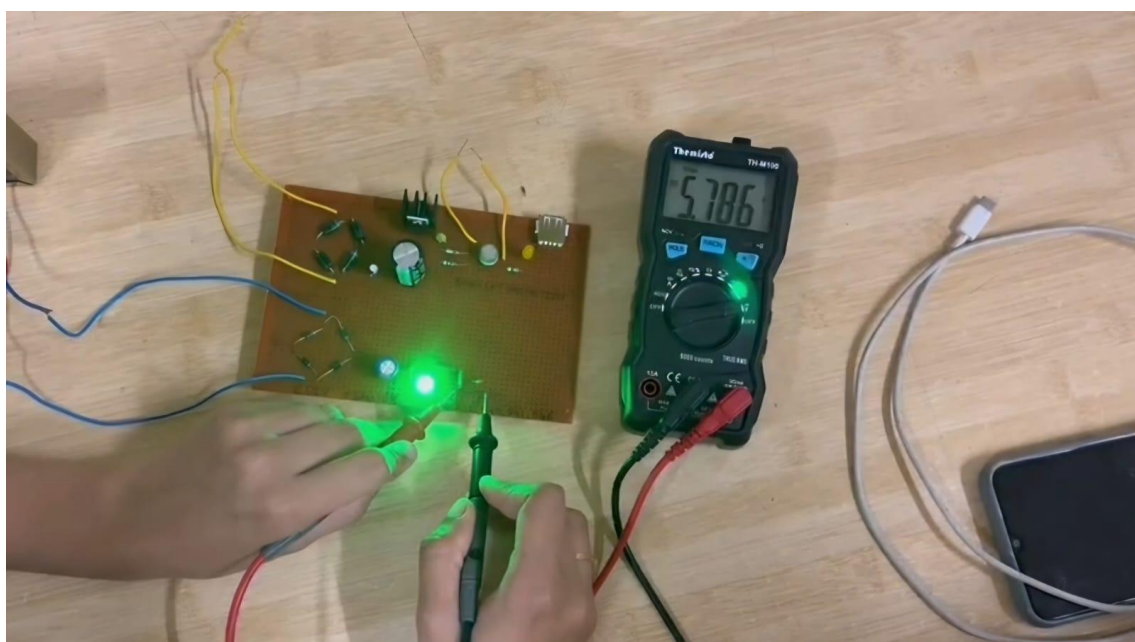


Fig: Measurement of output voltage of the charger with a digital multimeter

As presented by the results of the experiments, it is apparent that the mobile phone charger based on the IC can charge a real mobile phone with reasonable reliability. The required level of the output voltage of 5 V is kept during the entire charging period, which is crucial to eliminate battery damage and keep users safe.

Chapter 7

Conclusion and Reflections

7.1 Conclusion

A working mobile phone charger, rated at 5V and 1A, emerged clearly through use of the Lm323T linear voltage regulator. While delivering stable output under varying input conditions, it also handled changes in connected devices smoothly. Because the chip safeguards like automatic current control and heat based shutoff, reliability improved noticeably. When measured against a basic Zener-diode setup, performance gaps became obvious- especially under heavy demand where precision matters. That older method struggles when steady, strong current is needed, unlike the integrated solution which holds its response tighter. Experience from building this small system brought insights into real-world challenges: managing warmth across parts stood out, just as much as choosing elements carefully did. Simulating behaviour before physical trials revealed flaws early; later checks confirmed operation matched expectations closely.

7.2 Future scope

The current work can be expanded by implementing additional safety mechanisms and monitoring systems which include output over-voltage protection and input surge defense and output current monitoring systems. The charger can be integrated into a compact PCB with USB-A or USB-C output connectors to make it closer to regulators and buck converters to achieve better efficiency in applications that require minimal power waste and thermal management.

7.3 Application in the societal context

People need trustworthy mobile phone chargers because they use their phones for daily communication and learning activities and contact emergency services. The mobile phone battery and internal circuitry receive protection from a 5V power supply. The power supply prevents both over-voltage and over-current conditions which results in improved safety measures and extended device lifespan. Students who learn to design and understand these chargers will develop skills that enable them to create consumer electronics which improve safety standards and reduce energy consumption.

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MINI PROJECT 274

ORIGINALITY REPORT

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